

Rajdeep Gill

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Education

University of Manitoba

Bachelor of Science in Computer Engineering

May 2026

4.0 GPA

Experience

Robinhood

Software Engineer Intern

May 2025 - Sept 2025

Toronto, ON

- Reduced customer margin calls by **24%** and boosted trading activity by **13%** through a real-time maintenance dashboard built with **React and TypeScript**, featuring embedded trade execution and risk indicators.
- Architected a cross-platform margin experience using a **server-driven UI approach with gRPC** to unify business logic across Web, iOS, and Android, driving a **43%** increase in margin usage.
- Accelerated the adoption of a new **Protobuf design system** by implementing its foundational components in **Python and TypeScript**, and adding a **logging framework** to improve observability.
- Collaborated with 4 mobile engineers and led the redesign of the margin call experience on web by incorporating the maintenance dashboard into the existing margin call flow.
- Repaired the core integration suite for the margin experience using **Jest and Pytest**, adding **20+ test suites** with comprehensive mocking strategies, achieving **86%** coverage.

Biosensors Research Lab

Machine Learning Research Intern

May 2024 - Dec 2024

Winnipeg, MB

- Built a custom **CNN** achieving **85% accuracy** on 10,000+ diffraction images, providing a scalable, **non-invasive alternative** to trypan blue staining (**90-95% accuracy**) for assessing cell health.
- Built a **model validation GUI** in **Python** that displayed **SHAP-based feature attributions**, allowing biologists to debug predictions cell by cell
- Engineered an end-to-end **parallelized data pipeline** that improved morphological parameter extraction by **17%** and halved preprocessing runtime.
- Deployed a **centralized MySQL database** to replace manual hard-drive sharing, eliminating data loss and versioning conflicts while enabling researchers to run **cross-experiment queries** and collaborate on shared datasets.

Biosensors Research Lab

Machine Learning Research Intern

May 2023 - Aug 2023

Winnipeg, MB

- Engineered a **Python control framework** for lab oscilloscopes that abstracted low-level hardware commands into a high-level API, allowing **automation of entire experimental sequences** and cutting manual setup time.
- Slashed data analysis runtime by over **90%** (1 hour to <6 minutes) by refactoring legacy Python scripts, replacing inefficient loops with vectorized NumPy operations
- Developed a novel classification system using a Gaussian Mixture Model to determine cell viability from time-series impedance data, creating a probabilistic tool for assessing sample health

Projects

Void - AI App Builder (Next.js, TypeScript, Prisma, Docker)

- Developed an app to generate and deploy web apps from user prompts, integrating a usage-based Stripe billing system to monetize user compute time and API token consumption
- Implemented a persistent background job queue to process long-running AI build tasks asynchronously, allowing users to submit jobs and receive the completed result at a later time, even after closing the browser
- Utilized Docker for user code sandboxing, secure code execution and real-time application previews

Distributed Image Processing (C++, OpenMPI)

- Accelerated the processing of gigapixel-scale images by 3.8x compared to a single-threaded approach by designing an OpenMPI pipeline that partitioned data across a 4-node cluster
- Implemented a custom communication protocol to optimize data transfer latency and throughput.
- Utilized OpenMPI to handle the distribution of tasks across multiple nodes, ensuring scalability and reliability.

Skills

Programming Languages: C++, Python, TypeScript, JavaScript, Java

Frameworks and Libraries: React, NextJS, PyTorch, OpenMPI, Node.js, Express, Flask

Tools: Git, Docker, PostgreSQL, Github Actions, Bazel, gRPC, Protocol Buffers (Protobuf)